

Logging_Monitoring_Standard

Technical Standard

保险公司 技术开发部
内部文档 · 仅供授权人员查阅

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Logging & Monitoring Standard

> Reference: Google SRE Practices, Alibaba Cloud SLS Best Practices, ELK Community Standards, Prometheus Best Practices

1. Overview

1.1 Purpose

Standardize logging and monitoring alert practices to ensure comprehensive observability coverage, supporting rapid fault localization, business metric monitoring, and capacity planning.

1.2 Scope

This standard applies to all internally developed systems' logging, metrics collection, and alert rule definitions within the Technology Development Department.

1.3 Terminology

Term	Definition
Logging	Discrete event records
Metrics	Aggregatable numerical indicators
Tracing	Request-level trace tracking
Observability	Logging + Metrics + Tracing combined
Instrumentation	Embedding log/metric collection points in code

2. Logging Standards

2.1 Log Level Definitions

Level	Usage	Example	Handling
TRACE	Development debugging details	SQL parameters, loop intermediate values	Development only
DEBUG	Diagnostic information	Request params/results, state transitions	Off by default, on-demand
INFO	Key business events	Policy issuance, payment callback, task start/stop	Written to log files, long-term retention
WARN	Unexpected but recoverable	Retry triggered, rate limiting, config degradation	Written to files, triggers alert
ERROR	Feature unavailable / data error	DB connection failure, interface error, accounting imbalance	Immediate alert, on-call handling
FATAL	System crash	OOM, main thread exit	Immediate response, restart

2.2 Log Format Standards

Text Log Format:

```
{%d{yyyy-MM-dd HH:mm:ss.SSS}} [%thread] [%-5level] [%logger{36}] [%X{traceId}] - %msg%n
```

Example Output:

```
[2024-06-15 10:30:15.123] [http-nio-8080-exec-1] [INFO ] [c.c.i.underwriting.PolicyService] [t-abc123] - Policy issued successfully, policyNo=P2024000001
```

JSON Log Format (recommended for production):

```
{
  "timestamp": "2024-06-15T10:30:15.123+08:00",
  "level": "INFO",
  "thread": "http-nio-8080-exec-1",
  "logger": "c.c.i.underwriting.PolicyService",
  "traceId": "t-abc123",
  "message": "Policy issued successfully",
  "context": {
    "policyNo": "P2024000001",
    "amount": 1000.00,
    "duration": 152
  }
}
```

2.3 Key Instrumentation Points

Domain	Event	Level	Required Fields
Underwriting	Proposal submitted	INFO	proposalNo, productCode, channelCode
Underwriting	Decision completed	INFO	proposalNo, decision, reason
Policy Issuance	Policy issued success	INFO	policyNo, premium, effectiveDate
Policy Issuance	Policy issued failure	ERROR	policyNo, errorCode, errorMessage
Claims	Claim reported	INFO	claimNo, policyNo, incidentType
Claims	Claim settled	INFO	claimNo, payoutAmount, closeReason
Payment	Payment request sent	INFO	paymentId, amount, channel
Payment	Payment callback received	INFO	paymentId, status, channelOrderNo
Payment	Reconciliation mismatch	ERROR	paymentId, systemAmount, channelAmount
Renewal	Renewal reminder sent	INFO	policyNo, remindDate, channel
Renewal	Renewal payment result	INFO	policyNo, result, failReason
Policy Servicing	Endorsement received	INFO	endorsementNo, changeType, policyNo

Domain	Event	Level	Required Fields
Policy Servicing	Endorsement applied	INFO	endorsementNo, status, effectiveDate

2.4 Logging Governance Rules

- **No sensitive information** in logs: ID numbers, phone numbers, bank cards, passwords, tokens must be masked
- **No logging inside loops**: aggregate first, then log
- **No logging ERROR then rethrowing**: avoid duplicate logging
- **Exceptions must include full stack trace**: `log.error("msg", exception)`
- **Every log entry must include traceId**: for full trace correlation

2.5 Data Masking Rules

Data Type	Masking Method	Example
ID Number	Keep first 6, last 4	110101*****1234
Phone Number	Keep first 3, last 4	138****5678
Bank Card	Keep first 4, last 4	6222*****1234
Password/Token	Full mask	*****
Name	Keep surname	J***

3. Monitoring Standards

3.1 Four-Tier Monitoring

Layer	Metrics	Goal
Infrastructure	CPU, Memory, Disk, Network, GC	Resource levels, capacity planning
Application	QPS, RT, Error rate, Thread pool	Service quality (SLO)
Business	Issuance volume, Premium income, Payment success rate	Business health
UX	Page load time, API TTFB	User experience

3.2 Core Metrics Definition

RED Metrics (Rate / Errors / Duration):

Metric	Definition	Aggregation	Alert Threshold
Rate	Requests per second (QPS)	Sum	Exceeds 80% capacity
Errors	Error count/rate	Sum / Rate	Error rate exceeds 1%
Duration	Response time P50/P95/P99	Histogram	P99 exceeds 1000ms

USE Metrics (Utilization / Saturation / Errors):

Metric	Definition	Applicable To
Utilization	Resource utilization	CPU / Memory / Disk / Connection pool
Saturation	Queue length	Thread pool / Connection pool / Queue
Errors	Error count	All resources

3.3 Business Monitoring Metrics

Metric Name	Description	Aggregation	P1 Alert Threshold
policy_issued_total	Total policies issued	COUNT	Zero for 5 consecutive minutes
premium_total	Total premium income	SUM	50% drop vs same time yesterday
payment_success_rate	Payment success rate	rate(success)/rate(total)	Below 99%
underwriting_completion_time	Underwriting completion time	HISTOGRAM	P99 exceeds 30s
claim_settlement_time	Claim settlement time	HISTOGRAM	P99 exceeds 72h

3.4 Metrics Naming Convention

{namespace}_{component}_{metric_name}_{unit}

Examples:

insurance_underwriting_qps_total

insurance_payment_duration_seconds

insurance_core_db_connections_active

- Use underscore separators
- Unit at the end: _seconds, _bytes, _total
- All metrics include labels: service, environment, instance

4. Alert Rules

4.1 Alert Severity Levels

Level	Definition	Response Time	Notification
P0 (Critical)	Core feature completely unavailable / financial loss	5 minutes	Phone + SMS + IM
P1 (Severe)	Core feature impaired / widespread slowness	15 minutes	SMS + IM

Level	Definition	Response Time	Notification
P2 (Warning)	Non-core feature abnormal / capacity warning	1 hour	IM
P3 (Info)	Non-business anomaly / trend to watch	24 hours	Email

4.2 Alert Rule Template

```
# PrometheusRule example
groups:
- name: insurance.core.rules
rules:
# P0: Core service down
- alert: CoreServiceDown
expr: up{service="policy-service"} == 0
for: 1m
labels:
severity: P0
annotations:
summary: "Policy service is down"
description: "Instance {{ $labels.instance }} has been down for 1 minute"
# P1: High error rate
- alert: HighErrorRate
expr: rate(http_requests_total{status=~"5.."}[5m]) / rate(http_requests_total[5m]) > 0.01
for: 5m
labels:
severity: P1
# P2: Payment success rate drop
- alert: PaymentSuccessRateDrop
expr: payment_success_rate < 0.99
for: 10m
labels:
severity: P2
# P3: Slow query increase
- alert: SlowQueryIncrease
expr: rate(mysql_slow_queries_total[15m]) > 10
for: 15m
labels:
severity: P3
```

4.3 Alert Governance

- Do not create alert rules without thresholds
- Every alert must have a corresponding Runbook
- Weekly alert review to clean up ineffective rules
- Alert silences must have an expiry time; permanent silence is prohibited
- Alert notifications must be tied to the on-call schedule

5. Appendix

5.1 Reference Documents

- Google SRE Book
- Prometheus Best Practices

- OpenTelemetry Specification
- *Coding Standards* (Logging section)

5.2 Tool Stack

Domain	Recommended Tools	Alternatives
Log collection	Filebeat / Fluentd	Logstash
Log storage	Elasticsearch / Loki	ClickHouse
Log analysis	Kibana / Grafana Explore	—
Metrics collection	Prometheus / VictoriaMetrics	—
Metrics visualization	Grafana	—
Distributed tracing	Jaeger / Zipkin	SkyWalking
Alert management	AlertManager / Grafana Alerting	—
On-call scheduling	OnCall (PagerDuty)	—

5.3 Log Keyword Alert Rules

Keyword	Severity	Description
OutOfMemoryError	P0	OOM requires immediate response
Connection refused	P1	Service connection failure
Deadlock	P1	Deadlock causing service unavailability
Timeout	P2	Timeout anomaly requires attention
Retry exhausted	P2	Manual intervention needed
Circuit breaker opened	P2	Circuit breaker tripped
Degraded	P3	Service degradation, log for record